

Donghwa Kim

Full-Stack Engineer (React / Node.js / Realtime Systems)

Full-Stack Engineer with 4+ years of experience building scalable web applications across frontend, backend, and cloud infrastructure.

Strong experience with React, Next.js, and Node.js (NestJS), including real-time systems using WebSocket/WebRTC and production deployment on AWS.

Proven ability to design and deliver production-ready systems under real-world constraints (low connectivity, hardware integration), with a focus on performance, reliability, and user-facing products.

- Email: devmemorydh@gmail.com
 - GitHub: <https://github.com/devmemory>
 - Portfolio: <https://devmemory.github.io/#!/portfolio>
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TECHNICAL SKILLS

Frontend

React, Next.js, TypeScript, Zustand, Tailwind

Backend

Node.js, NestJS, REST APIs, PostgreSQL, Redis, TypeORM

Infrastructure / DevOps

AWS (EC2, S3, Route53), Docker, NGINX, GitHub Actions

Realtime / Media Systems

WebRTC, WebSocket, HLS, FFmpeg

EXPERIENCE

Outloud Company — Lead Full-Stack Engineer

Aug 2025 – Jan 2026

Led end-to-end development of an offline-first music booth platform integrating real-time media processing and hardware devices.

- Re-architected prototype into a scalable, production-ready system deployed in physical stores
- Designed and implemented PostgreSQL schema from scratch (ERD, relationships, indexing)

- Built REST APIs using NestJS for media workflows and device communication
- Implemented presigned URL-based architecture to separate server and storage, reducing latency during media uploads
- Designed Electron-based system combining web UI with native hardware control
- Built media processing pipeline using FFmpeg and HLS streaming
- Integrated external devices (payment terminal, camera, printer) via SDKs and browser APIs
- Established CI/CD pipelines using AWS, Docker, and GitHub Actions
- Built internal operational tools using Webhooks and Google Sheets

Impact

- Delivered MVP in 4 weeks, full production system in 4 months
- Successfully launched in real stores and contributed to revenue generation
- Improved system responsiveness and reduced server load under heavy media usage
- Reduced incident response time through improved deployment and monitoring

Luxrobo — Frontend Engineer (Realtime Systems)

Mar 2024 – Apr 2025

Developed real-time systems and frontend architecture for AI circuit design and smart parking platforms.

- Led frontend architecture design and implementation
- Collaborated on database schema design for dashboard features
- Built 3D PCB viewer using Three.js and circuit editor using JointJS
- Developed internal tooling for GraphQL code generation, improving development efficiency
- Implemented real-time CCTV emergency communication system using WebRTC
- Worked closely with backend engineers to align API and real-time communication design
- Set up CI/CD pipelines using Docker, NGINX, and GitHub Actions

Impact

- Delivered investor-facing AI platform prototype
- Reduced delivery timeline by 2 weeks, enabling faster go-to-market

Gaon Platform — Frontend Engineer

Mar 2023 – Sep 2023

Improved performance and stability of an AI-based monitoring platform under tight deadlines.

- Refactored frontend architecture into modular and scalable structure
- Migrated polling system to WebSocket-based real-time communication
- Designed authentication and authorization system (JWT, role-based access)
- Built data visualization dashboards using D3.js and Highcharts

Impact

- Delivered project on time despite initial delays
- Improved system performance, scalability, and maintainability

Additional Project (2 weeks)

- Built pump failure diagnostic visualization tool independently
 - Delivered 1 week ahead of schedule
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Lexynaptics — Frontend Engineer

Jun 2022 – Dec 2022

Built MVP for real-time legal consultation platform.

- Designed frontend architecture for web and mobile
- Implemented 1:1 video communication using WebRTC
- Integrated payment, identity verification, and third-party APIs
- Built user analytics and event tracking system

Impact

- Delivered MVP before investor meetings
 - Enabled successful investment preparation
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Reindeer — Mobile App Developer

Apr 2021 – Jun 2022

Worked on travel guide and reservation platform.

- Refactored legacy Flutter codebase and improved performance
- Built custom map solution to overcome SDK limitations
- Integrated payment systems and external APIs
- Improved UI/UX and implemented new design system

Impact

- Improved rendering performance by 10+ FPS
 - Reduced crash rate by ~90%
 - Stabilized service for continued operation and expansion
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PROJECTS

PCB 3D Viewer

<https://devmemory.github.io/gerber-3d-viewer>

- Built 3D visualization tool for PCB Gerber files using Three.js
 - Implemented CI/CD pipeline for automated deployment
 - Received collaboration proposal from PCBWay
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Cursor Ubuntu Installer

<https://github.com/devmemory/cursor-ubuntu-installer>

- Built shell-based installer for Cursor IDE on Ubuntu
 - Implemented automated version update system
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KakaoMap Flutter Package

https://pub.dev/packages/kakaomap_webview

- Developed and published Flutter package for Kakao Map integration
 - Used in 100+ applications
 - Achieved 27 likes on Pub.dev
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EDUCATION

Kongju National University

B.S. in Electrical, Electronics & Control Engineering

(2010 – 2017)